

CVM UNIVERSITY
B.Sc CA&IT
Semester – IV
Relational Database Management Systems - II

Assignment 01 : Table creation with constraints:

1. Create table CUSTOMER_MST and insert following records.

Field Name	Type	Size	Constraints
CUST_NO	VARCHAR2	6	Must start with ANA or VVN or BAK, PRIMARY KEY
CUST_NAME	VARCHAR2	20	Can not hold NULL values
ADDRESS	VARCHAR2	20	
CITY	VARCHAR2	10	Can hold values ANAND, V V NAGAR or BAKROL

CUST_NO	CUST_NAME	ADDRESS	CITY
ANA001	ANUJ PATEL	PATEL STREET	ANAND
VVN001	SURAJ SHAH	NANA BAZAR	V V NAGAR
ANA002	BINITA THAKKAR	LAKSH APT.	ANAND
BAK001	ANIL PATEL	SUNRISE SOCIETY	BAKROL
VVN002	VIJAY RAY	MOTA BAZAR	V V NAGAR
ANA003	ROMA BHATT	TOWN HALL ROAD	ANAND

2. Create table CUSTOMER_DTL and insert following records.

Field Name	Type	Size	Constraints
CUST_NO	VARCHAR2	6	Reference to CUSTOMER_MST table
BILL_MON	VARCHAR2	3	
PREV_READ	NUMBER	5	
CURR_READ	NUMBER	5	
UNIT_PRICE	NUMBER	5,2	MUST BE > 0
DUE_DATE	DATE		

CUST_NO	BILL_MON	PREV_READ	CURR_READ	UNIT_PRICE	DUE_DATE
ANA001	NOV	1200	2500	5.50	26-DEC-20
VVN001	DEC	275	365	5.50	20-JAN-20
ANA002	NOV	145	570	5.50	26-DEC-20
BAK001	DEC	45	145	5.50	20-JAN-20

VVN002	DEC	600	680	5.50	20-JAN-20
ANA003	NOV	2500	2650	5.50	26-DEC-20

3. Create table DOCTOR_MASTER and insert following records.

Field Name	Type	Size	Constraints
DOCID	VARCHAR2	3	PRIMARY KEY
DNM	VARCHAR2	15	
CASE_FEE	NUMBER	3	

<u>DOCID</u>	<u>DNM</u>	<u>CASE_FEE</u>
D01	Viraj Patel	300
D05	Mansi Joshi	250
D03	Alok Sinha	400
D02	Roshni Bhatt	500
D04	Kerav Raj	200

4. Create table PATIENT_DETAIL and insert following records.

Field Name	Type	Size	Constraints
PID	NUMBER	2	PRIMARY KEY
PNM	VARCHAR2	20	References to DOCTOR_MASTER
DOCID	VARCHAR2	3	
CONS_FEE	NUMBER	3	
MEDI_AMT	NUMBER	3	
TEST_FEE	NUMBER	4	

PID	PNM	DOCID	CONS_FEE	MEDI_AMT	TEST_FEE
1	Purav	D01	100	300	0
2	Sonu	D02	150	700	3000
1	Purav	D05	80	200	500
4	Meha	D03	200	110	0
1	Purav	D05	50	650	200
3	Niraj	D03	150	800	1000
2	Sonu	D03	175	350	0
3	Niraj	D02	75	150	0

Assignment 02 : PL/SQL Block

1. Write a PL/SQL block to print "HELLO WORLD".
2. Write a PL/SQL block that will take two integer numbers. Print addition, subtraction, multiplication, division of the inputted numbers.
3. Write a PL/SQL block to take an input in the form of number and check whether the number is EVEN or ODD.
4. Write a PL/SQL block to take an input in the form of number and check whether the number is positive, negative or zero.
5. Write a PL/SQL block to display first 5 natural numbers. (Use LOOP..END LOOP)
6. Write a PL/SQL block to display first n natural numbers. (Use WHILE..LOOP)
7. Write a PL/SQL block to display first n natural numbers and sum of first n natural numbers. (Use FOR LOOP)
8. Write a PL/SQL block to display first n natural numbers in reverse order.

Assignment 03 : CURSOR

1. Write a cursor to display employee number, employee name, salary, job and department number of all the employees using %rowtype.
2. Write a cursor to input job form the user and display employee information for the inputted job.

Employee Information

Job : CLERK

Employee Number	Employee Name	Salary	Commission	Hiredate	Department Number
**	**	**	**	**	**

Total Salary: ****

3. Write a cursor to display electricity bill report in the following format:

Electricity Bill Report

Customer Number	Customer Name	Previous Readings	Current Reading	Consumption	Unit	Total Price	Bill
*	**	**	**	**	**	***	
*	**	**	**	**	**	***	

- Write a cursor to display Department wise Employee information in following format.

```

-----
Department wise Employee Information
-----
Department No. : XX      Department Name: XXXX
-----
Employee   Employee   Job   Salary   Commission   Hiredate
Number     Name
-----
**         **         **    **     **           **
-----
Department No. : XX      Department Name: XXXX
-----
Employee   Employee   Job   Salary   Commission   Hiredate
Number     Name
-----
**         **         **    **     **           **
-----

```

- Write a cursor to display doctor wise patient report in following format.

```

-----
DOCTORWISE PATIENT REPORT
-----
Doctor ID: D01   Doctor: Viraj Patel   Case fee: 300
Patient ID   Patient Name   Consulting fee   Medicine amount   Test fee   Total amount
-----
1           Purav         100             300               0          400
.....

```

Total amount = Consulting fee + Medicine amount + Test fee

Assignment 04 : FUNCTION

- Create a function to calculate simple interest.
- Create a function to accept department number and return department name.
- Create a function to accept employee salary and department number and calculate bonus.

Rules to calculate bonus:

If Department number=10 then Bonus=5% of salary

If Department number=20 then Bonus=10% of salary

If Department number=30 then Bonus=15% of salary

Else Bonus=20% of salary

Assignment 05 : PROCEDURE

1. Write a procedure to accept two numbers and return summation and subtraction of two numbers.
2. Write a procedure to accept marks of 3 subjects and return total and percentage.
3. Write a procedure to accept employee number. Return employee name and job of the employee.

Assignment 06 : PL/SQL Block with exception handling. (EMP and DEPT table)

1. Write a PL/SQL block to take an input in the form of department number. Print department information if the department number is present in the DEPT table else print appropriate message and terminate the block.
2. Write a PL/SQL block to take an input in the form of job. Print employee information if the job is with one employee in the EMP table. If the job is with more than one employee print appropriate message and terminate the block, and if the job is not present print appropriate message and terminate the block.